Hierarchical federated learning workflow

Local training | Hierarchical aggregation | Few-shot infereence

Battery data: LFP (HUST / MIT) | NCA (TJU) | NCM (TJU / XJTU)

303 source cells / 12 clients | 75 target cells / 5 domains | Simulated aggregation

A FEDERATED TRAINING

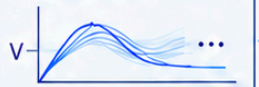
45 rounds

1 **Broadcast shared encoder and domain heads**

2 **Local training: two epochs**

Shared | Domain head

Charge-record input



17-D features

ACfC encoder



17-32 | CfC x2 | 36

Domain head



MLP: 36-32-1

SOH prediction

$\hat{y}$



Huber loss

Paired supervision

Backprop.

Next round

3 **Upload updates (simulated pairwise masking)**

4 **Aggregate shared modules**

Within domain: battery-count weights

Within material: equal sources; then equal materials

4 **Aggregate domain heads**

Battery-count weighted within each domain

After base training: fit source-specific mappings with the encoder frozen

Curve heads / reference mappings / residuals / experts

B TARGET ADAPTATION & INFERENCE

Early support records

20 labelled observations

Freeze shared encoder

Adapt the domain head

Local SOH inference

Adapted head + fixed source mappings

Process branches & fuse

Filters / residuals / routing

Equal mean of 3 complete outputs

Repeat the full target pipeline for each saved model; average only the three final outputs.

Shared: ACfC encoder | Personalized: domain heads and source-specific mappings